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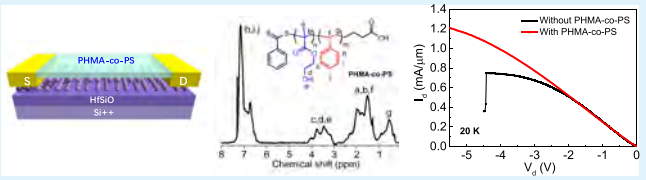
Performance and Reliability Improvement under High Current Densities in Black Phosphorus Transistors by Interface Engineering

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Supporting Information

ABSTRACT: Few-layer black phosphorus (BP) has recently emerged as a promising two-dimensional (2D) material for electronic and optoelectronic devices because of its high mobility and tunable band gap. However, BP is known to quickly degrade and oxidize in ambient conditions by breaking of the P−P bonds. As a result, there is a growing need to encapsulate BP that avoids oxygen and water while retaining the high electric performance of the devices. Here, we demonstrate a hydrophobic polymer encapsulation technique with improved thermal conductivity for high current density, which preserves the electrical properties of BP back-gate transistors compared to the commonly used Al_2O_3 encapsulation with improved mobility and minimal traps. The on−off ratio increases by more than an order of magnitude at room temperature and more than 4 orders of magnitude at cryogenic temperatures. High field transport shows the first systematic study on unprecedented breakdown characteristics up to −5.5 V for the 0.16 μm transistors with a high current of 1.2 mA/μm at 20 K. These discoveries open up a new way to achieve high-performance 2D semiconductors with significantly improved breakdown voltage, on−off ratios, and stability under ambient conditions for practical applications in electronic and optoelectronic devices.



KEYWORDS: *black phosphorus, field-effect transistors, polymer, encapsulation, high-field transport, breakdown voltage, reliability*

INTRODUCTION

Following Moore's law, the device scaling of Si complementary metal-oxide semiconductor technology is reaching the fundamental physical limits.[1] As one of the most promising candidates for beyond-Si novel channel materials, two-dimensional (2D) materials have the best immunity to the short channel effects due to the ultrathin body nature.[2] Thus, 2D layered materials hold great potential for low-power, high-performance electronic applications.[3] In the past decade, graphene has attracted much attention due to its extraordinarily high carrier mobility and saturation velocity. However, transistors based on graphene cannot be switched off due to a zero band gap, which severely limits its applications in electronic devices.[2] Although many efforts have been made to open a band gap in graphene, it still remains a great challenge. Recently, semiconducting transition-metal dichalcogenides like MoS_2 have been receiving substantial attention owing to the presence of a reasonable band gap (1.2−1.8 eV) and suitable electrical properties.[4] Demonstrations on various applications have been made on MoS_2, such as logic transistors,[4−7] radio-frequency transistors,[8] photodetector,[9] sensors,[10] flash memory,[11] logic circuits, and amplifiers.[12] However, from a practical point of view, the performance of MoS_2 devices is yet to improve due to the relatively low carrier mobility.[13,14]

As an emerging 2D semiconductor, layered black phosphorus (BP) has a finite direct band gap of 0.3 eV for bulk and 2 eV for monolayer, with a high carrier mobility of around 1000 $cm^2/V/s$. High-performance p-type field-effect transistors (FETs) based on few-layer BP have been reported,[15−21] as well as applications in radio-frequency transistors and photodetectors.[22−25] Moreover, anisotropic electric conductance,[26] thermal conductivity,[27,28] and optical responses have also been demonstrated experimentally,[29,30] providing many opportunities for novel electronic and optoelectronic devices applications. To date, most studies have been focused on large devices and low-field transport properties of BP devices, where the output current is far below the standard silicon transistors.[17,31,32] However, electrical transport under high field is crucial for any practical application. It remains a challenge to obtain a high output current while maintaining a high on/off ratio for BP transistors, and viable methods are yet to be explored. Moreover, the instability of BP in an environment with oxygen and moisture is another serious issue for practical applications.[33,34] Previous studies have shown that the introduction of AlO_x encapsulation layer could protect the surface of BP and improve the stability; however, this

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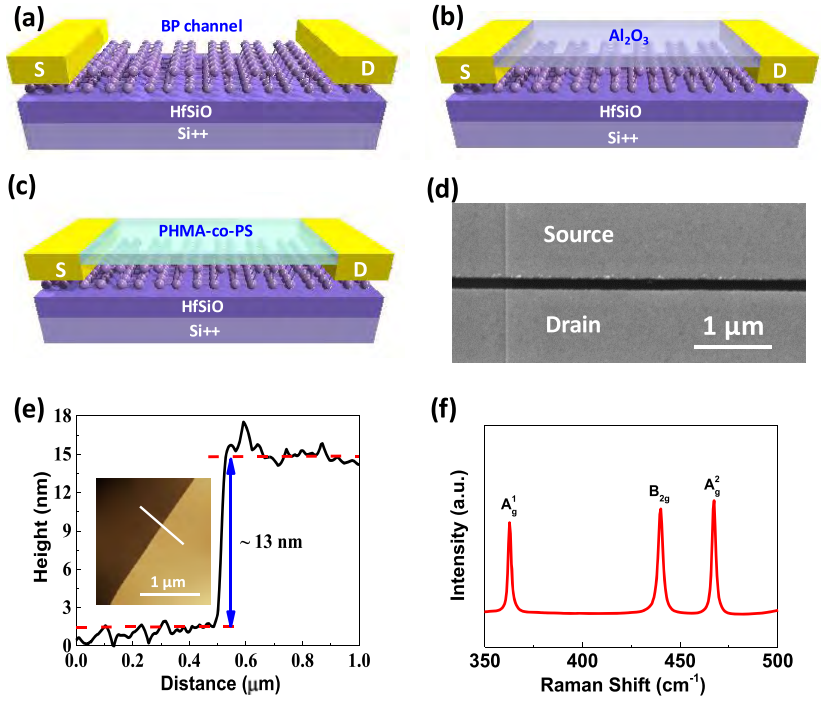


Figure 1. Schematic of the few-layer BP devices before (a) and after Al_2O_3 (b) and PHMA-*co*-PS (c) encapsulation. S: source. D: drain. (d) An SEM image of a BP backgate transistor with a 0.16 μm channel length. (e) The thickness of a BP flake along the white line. The scale bar is 1 μm. Inset of (e), AFM image of the BP flake. (f) Raman spectra collected from the BP flake on the HfSiO/Si substrate.

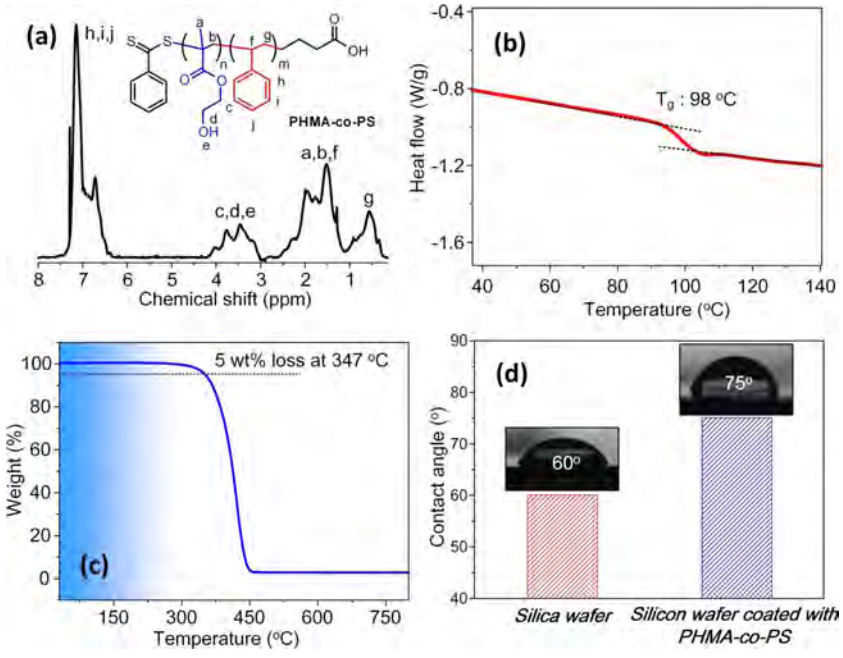


Figure 2. (a) NMR spectra, (b) DSC, and (c) TGA curves of PHMA-*co*-PS. (d) Contact angle for silica wafer (left side) and silicon wafer coated with PHMA-*co*-PS (right side).

improvement is at the cost of severe performance degradation with reduced drain current up to 30 times.[35−38] This is likely due to the introduction of moisture during atomic layer deposited (ALD) process and charge traps in the dielectrics, as well as the formation of PO_x layer.[39] The ability to minimize the performance degradation is of vital importance for high-performance BP devices.

In this work, we have developed a novel passivation method using poly(2-hydroxyethyl methacrylate)-*co*-poly(styrene) (PHMA-*co*-PS) for BP transistor, with greatly improved electrical transport properties at electric high field. With the effective PHAM-co-PS encapsulation, the off current as well as improved on/off ratio about 4−10 times can be achieved in these BP FETs with PHMA-*co*-PS polymer as well as an increase in the breakdown voltage. The maximum drain current of BP FETs with PHMA-*co*-PS encapsulation is up to 1.2 mA/μm at 20 K. These results open up a new way to achieve high-performance 2D semiconductors with greatly

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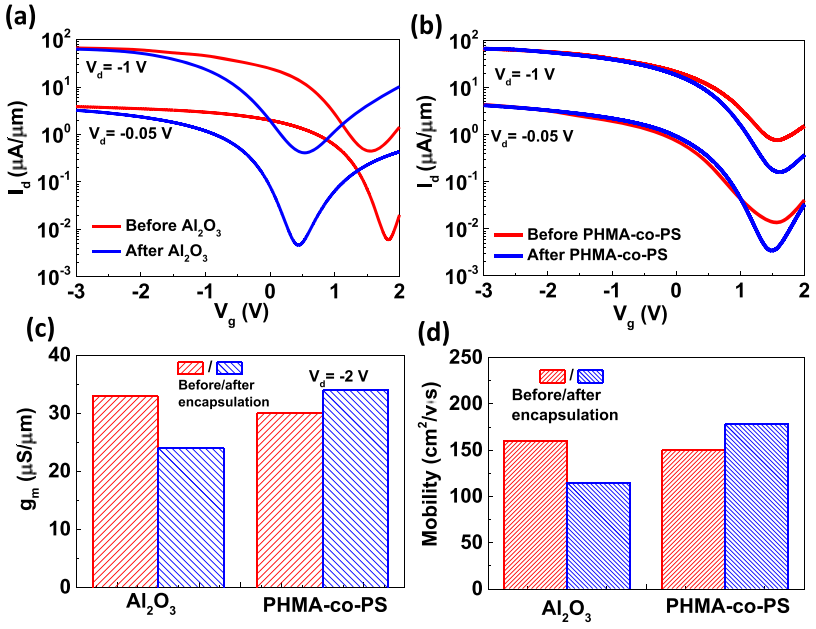


Figure 3. Comparison of transfer characteristics of the 2 μm BP transistors before and after (a) Al_2O_3 and (b) PHMA-*co*-PS encapsulation at 300 K. (C) Comparison of g_m of the devices before and after encapsulation at $V_d = -2$ V at 300 K. (d) The corresponding mobility is compared.

improved on−off ratios, breakdown voltage, and stability under ambient conditions for practical applications in electronics and optoelectronics.

RESULTS AND DISCUSSION

BP transistors with channel length from 0.16 to 2 μm are fabricated using standard nanofabrication on 24 nm HfSiO dielectric on Si substrate. Figure 1a shows a schematic of a fabricated BP FET with atomic layer deposited (ALD) HfSiO as gate dielectric. Figure 1b,c shows the schematic of BP FETs with Al_2O_3 and PHMA-*co*-PS encapsulation layers. Figure 1d shows the scanning electron microscopy (SEM) image of a BP device with 0.16 μm channel length. Figure 1e shows a representative exfoliated BP flake of 13 nm measured by atomic force microscope. The three Raman-activated modes ($A_{1g} \sim 361$ cm^{-1}, $B_{2g} \sim 438$ cm^{-1}, and $A_{2g} \sim 466$ cm^{-1}) in Raman spectra as shown in Figure 1f are consistent with previous work on BP characteristics.[40,41] After the fabrication of the devices, two materials were deposited to encapsulate the BP devices to improve their air stability. Al_2O_3 with a thickness of 20 nm was deposited by ALD at 150 °C using trimethylaluminum and water as precursors. PHMA-*co*-PS was spin-coated onto the measured devices at 6000 rpm and then prebaked at 175 °C for 5 min on a hot plate to remove the residual solvent. Details about the fabrication are given in Figures S2 and S3 in the Supporting Information. The thickness of PHMA-*co*-PS is approximately 20 nm, as determined by ellipsometry, which is also confirmed on BP by atomic force microscopy (AFM). Electrical measurements were carried out for the completed FETs in a Lakeshore probe station in vacuum (<10^{-5} Torr) using an Agilent B1500A semiconductor parameter analyzer from 300 to 20 K.

Poly(2-hydroxyethyl methacrylate) (PHMA) is a nontoxic hydrophilic polymer and its copolymers have been used as carriers for enzyme and protein immobilization, as absorbents for chromatographic applications, and as scavengers for removing metal ions from solution.[42,43] When PHMA is copolymerized with styrene (St), it should be possible to enhance the hydrophobicity and film-forming ability. In this study, we used reversible addition fragmentation chain transfer polymerization to synthesize well-defined PHMA-*co*-PS because the process is known to be tolerant to a wide range of monomers, functional groups, and reaction conditions.[44] The synthesized copolymer has a well-controlled molecular weight (M_n = 14 500 g/mol) and a low polydispersity (PDI = 1.26). Nuclear magnetic resonance (NMR) results in Figure 2a provide further information about the corresponding characteristic structures of the resulting copolymer, which indicates that the repeating units of PHMA to PS segment is about 1:3 by integral area calculation between hydroxyethyl groups of PHMA and aromatic ring of PS. The thermal properties of PHMA-*co*-PS were measured using differential scanning calorimetry (DSC) and thermogravimetric analyzer (TGA). The results in Figure 2b,c show that its glass transition temperature (T_g) and 5 wt % loss was 98 °C and 347 °C, respectively. Moreover, water contact angle test showed that the silicon wafer coated with PHMA-*co*-PS is more hydrophobic ($\theta = 75°$) compared to the pure silicon wafer ($\theta = 60°$) (Figure 2d). The good thermal stability and hydrophobicity of the resultant copolymer are ideal for surface passivation purpose for the ambient-sensitive black phosphorus material.

Because Al_2O_3 is the most commonly used dielectric for passivating the BP surface, we first compared the electrical characteristics of BP FETs with a 2 μm channel length before and after Al_2O_3 encapsulation at 300 K as shown in Figure 3a. It is clear that the transfer characteristics (I_d−V_g) curves with $V_d = -0.05$ and −1 V shift toward a negative voltage at about 1.5 V after Al_2O_3 encapsulation. This is consistent with previous studies and can be mainly attributed to the surface oxidation of BP by the introduction of moisture during H_2O-based Al_2O_3 process.[38,45]

Figure 3b compares the I_d−V_g curves of BP FETs before and after PHMA-*co*-PS encapsulation at $V_d = -0.05$ and −1 V. It is evident that the PHMA-*co*-PS encapsulation layer reduces the off current and increases the on/off ratio over four times. This could be due to the removal of charge-trapping adsorbates

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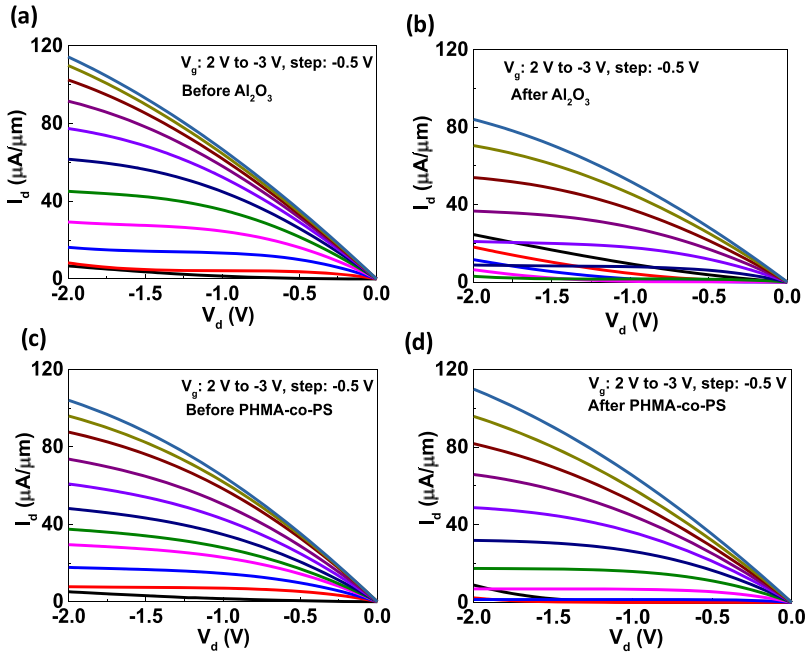


Figure 4. Comparison of output characteristics of the 2 μm BP transistors before and after encapsulation with (a, b) Al_2O_3 and (c, d) PHMA-*co*-PS at 300 K.

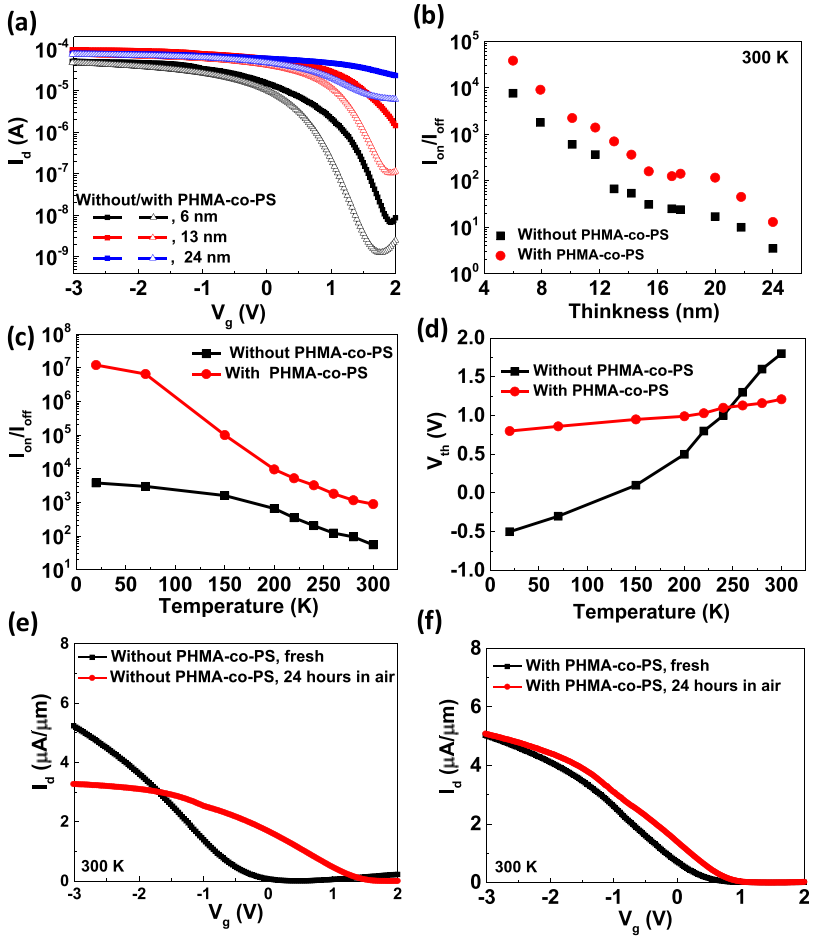


Figure 5. (a) Transfer characteristics of the 0.16 μm BP transistors with three different thicknesses before and after PHMA-*co*-PS encapsulation with $V_d = -0.05$ V at 300 K. (b) Comparison of the on/off ratios for encapsulated and unencapsulated BP FETs versus different BP flake thickness. Comparison of (c) on/off ratio and (d) V_{th} for 13 nm BP devices before and after PHMA-*co*-PS encapsulation from 300 to 20 K. Comparison of the transfer curves BP FETs with $V_d = -0.05$ V at 300 K in the ambient for (e) the unencapsulated device and (f) the encapsulated device.

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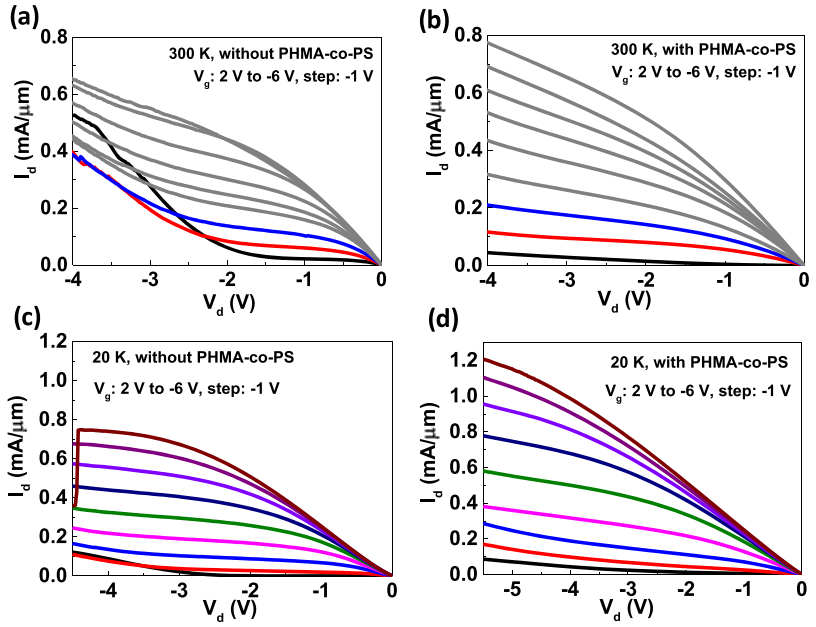


Figure 6. (a, b) Output characteristics of the unencapsulated and encapsulated BP devices with PHMA-*co*-PS at 300 K. V_d is from 0 to −4 V. (c) Breakdown characteristics of the unencapsulated BP device at 20 K. V_g is from 2 to −6 V in steps of −1 V and V_d is up to −4.5 V. (d) Breakdown characteristics of the encapsulated BP device at 20 K. V_d is up to −5.5 V.

H_2O and O_2 by encapsulation, which indicates that the adsorbates H_2O and O_2 on the surface of BP channel significantly affect the off current, and our polymer passivation opens a reliable way to reduce the off current and improves the on/off ratios. We extracted peak transconductance (g_m) at V_d = −2 V and field-effect mobility at V_d = −0.05 V before and after two different encapsulations, and the results are shown in Figure 3c,d. It is clear that both transconductance and mobility decrease after Al_2O_3 encapsulation, whereas both parameters improve after PHMA-*co*-PS encapsulation, showing the obvious advantage of the polymer encapsulation for electrical performance in BP transistors.

Figure 4a,b shows the output characteristics of BP transistors before and after Al_2O_3 encapsulation, respectively. It is evident that the drain current at V_d = −2 V decreases from 114 μA/μm before encapsulation to 85 μA/μm after encapsulation. This is consistent with the above results and the previous studies and is mainly attributed to the surface oxidation of BP during the H_2O-based Al_2O_3 process.[38,45] Figure 4c shows the I_d−V_d curves of BP FETs without PHMA-*co*-PS encapsulation and the maximum drain current is 104 μA/μm. After encapsulation, there is a slight increase in maximum drain current, as shown in Figure 4d. Importantly, the off current at V_d = −2 V reduces from 5.1 to 1.3 μA/μm, a reduction of up to four times. These results show that it is vital to passivate the BP surface using a polymer encapsulation layer for low-power and high-performance BP FETs.

To investigate the effect of flake thickness on the electrical properties, the I_d−V_g curves of BP FETs with three typical thickness at V_d = −0.05 V for the 0.16 μm channel length before and after PHMA-*co*-PS encapsulation are shown in Figure 5a. The BP devices with PHMA-*co*-PS indeed exhibit reduced off current and improved on/off ratios for all flakes. Figure 5b summaries the on/off ratios change before and after PHMA-*co*-PS encapsulation for various BP thicknesses at room temperature. The channel length is fixed at 0.16 μm. It is evident that the on/off ratio increases by a factor of 4−10 by a polymer coating to passivate the surface of BP, and this improvement is independent of the flake thickness. To obtain more in-depth insights into the impact of PHMA-*co*-PS encapsulation on the electrical characteristics of the BP devices, we compared the on/off ratios and V_{th} extracted from the transfer curves at various temperatures. Figure 5c shows the change in the on/off ratios for the 13 nm thick BP devices. On the one hand, the on/off ratios increase with decreasing temperature due to the reduced thermal emission current.[46] On the other hand, the on/off ratio of the device with PHMA-*co*-PS encapsulation is always larger than that of without encapsulation at all temperatures, approaching 10^7 at 20 K (Figure S4 in the Supporting Information). This indicates that the encapsulation layer is stable and reliable to reduce the off current of the BP devices. Figure 5d plots the threshold voltage (V_{th}) of BP FETs before and after encapsulation as a function of temperature. Before encapsulation, V_{th} decreases dramatically from 1.9 to −0.5 V as the temperature decreases from 300 to 20 K. However, the V_{th} remains almost the same after encapsulation in a range of 0.8−1.2 V. In view of the above results and the fact that coating BP FETs with PHMA-*co*-PS only inhibits the adsorption of moisture on the BP surface, we believe that the change in the on/off ratios and V_{th} is strongly related to moisture and oxygen molecule.[47] Although the BP transistors are measured in vacuum, the moisture and oxygen molecule absorbed in the channel surface cannot be removed by pumping in vacuum. The absorbed H_2O and O_2 on the BP surface act as charged impurities,[33,47,48] which leads to an increase in the off current and a decrease in the on/off ratios accordingly. Moreover, the absorbed molecules induce charge traps through either the capture or the emission of carriers during V_g swept and cause the instability of V_{th}. Therefore, passivation of the BP surface is of vital importance for high-performance BP FETs. The stability of 6 nm thick BP devices with and without PHMA-*co*-PS encapsulation in the air is also investigated and the electrical characteristics at 300 K are shown in Figure 5e,f. It is evident

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that the on-state current and the on/off ratios degrade drastically for the device without encapsulation. As expected, the encapsulated device showed better stability with almost unchanged electrical properties, suggesting excellent passivation for practical applications in ambient conditions.

It is important to evaluate and improve the device reliabilities for any technological relevant applications. However, few studies have focused on the high-field transport and breakdown phenomena in the BP transistors at short-channel devices so far.[49−51] Here, we present a comprehensive experimental characterization of the encapsulation effect on high-field transport for short-channel BP transistors, which is vital to understand the limiting mechanisms of the maximum current density of BP and gain insights under practical operating conditions. Figure 6a compares the I_d-V_d characteristics of the 0.16 μm BP device on the 13 nm flake before and after PHMA-*co*-PS encapsulation, with V_g changing from 2 to −6 V at drain voltage V_d reaching −4 V at 300 K. It is apparent that the off current increases dramatically up to 0.38 mA/μm when V_d is larger than −2 V, which poses a severe limitation for applications in low-power, high-performance technology. After polymer encapsulation, the off current shows very nice saturation behavior at large V_d, the impact ionization-like behavior is greatly suppressed with an order of magnitude smaller current than that before encapsulation at high drain bias. This shows that the encapsulation layer is effective in reducing the off current, which is consistent with the results in Figure 2. Figure 6c shows the output characteristics of the unencapsulated device at 20 K. The maximum drain current is 0.75 mA/μm at V_g = −6 V and V_d = −4.4 V and the off current decreases to 104 μA/μm at V_d = −4.4 V due to the reduced thermal emission current at low temperature. When the drain voltage increases to −4.5 V, we observed the breakdown of the BP transistor, wherein the current suddenly drops and can no longer recover. Figure 6d shows the well-behaved output characteristics of the PHMA-*co*-PS-encapsulated device with V_d up to −5.5 V. The on-state current is 1.07 mA/μm at V_g = −6 V and V_d = −4.4 V and the off current is 54 μA/μm at V_d = −4.4 V. Therefore, the PHMA-*co*-PS encapsulation results in 43% higher drain current and 58% lower off current. More importantly, the device did not show any degradation even at V_d = −5.5 V with no breakdown, and a high drain current of 1.2 mA/μm and a remarkably high current density of 9.2 MA/cm^2 are achieved.[49,51,52] Low off current is useful to reduce the standby power consumption and high on-state current can provide high switching speed. After encapsulation, the adsorption process is greatly suppressed and the formation of PO_x species is minimal, preserving the physical, electrical, and chemical integrity of BP. Meanwhile, the in-plane thermal conductivity of PHMA-*co*-PS ($\kappa \approx 0.2$ W/(m·K)) is 10 times higher than that of air in vacuum. Therefore, the encapsulation can increase thermal spreading and the breakdown voltage accordingly.[49] These results show that proper encapsulation is necessary to ensure high performance and low power of BP devices for nanoelectronics applications. Such studies are also essential to gain a deep understanding of the electrical properties for further progress toward ultrascaled device applications of 2D materials.

CONCLUSIONS

In conclusion, we have demonstrated high-performance BP transistors with low off current and high breakdown voltage by effective encapsulating. The on/off ratios improve 4−10 times and with greatly improved breakdown characteristics after PHMA-*co*-PS encapsulation. This is mainly because the encapsulation layer prevents BP from adsorbing moisture and oxygen molecular species to suppress the formation of PO_x species, as well as improved thermal conductivity at high current densities. Meanwhile, the PHAM-co-PS encapsulation technique exhibits excellent long-term stability in air and processes in a manner compatible with mainstream semiconductor devices manufacturing, which can be applied at wafer-scale. A high drain current of 1.2 mA/μm and a large on/off ratio around 10^7 are achieved at 20 K. This work not only offers new insights into the fundamental effect of encapsulation on electrical characteristics and breakdown voltage but also reveals the enormous potential of BP for technological applications.

METHODS

The few-layer BP were mechanically exfoliated from bulk crystal (Smart Elements) in an N_2-filled glovebox and then transferred on a p ++ Si substrate coated with 24 nm of atomic layer deposited HfSiO. The deposition of HfSiO film involves precursors of tetrakis-(ethylmethylamino)hafnium, tris(dimethylamino)silane, and O_3. Source/drain metal contacts were formed by e-beam lithography and 20/50 nm Ni/Au metal deposition and lift-off process. Electrical measurements were carried out in a probe station in vacuum ($<10^{-5}$ Torr) using an Agilent B1500A semiconductor parameter analyzer.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.8b16507.

SEM image of BP devices; atomic layer deposition of Al_2O_3; the fabrication of PHMA-*co*-PS polymer; effect of PHMA-*co*-PS polymer capping on on/off ratio of BP devices; effect of PHMA-*co*-PS polymer capping on breakdown voltage of BP devices; effect of PHMA-*co*-PS polymer and Al_2O_3 capping on the hysteresis of BP devices (PDF)

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Author Contributions

X.L. and J.W. contributed equally to this work.

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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REFERENCES

(1) del Alamo, J. A. Nanometre-Scale Electronics with III–V Compound Semiconductors. *Nature* **2011**, *479*, 317–323.
(2) Schwierz, F. Graphene Transistors. *Nat. Nanotechnol.* **2010**, *5*, 487–496.
(3) Fiori, G.; Bonaccorso, F.; Iannaccone, G.; Palacios, T.; Neumaier, D.; Seabaugh, A.; Banerjee, S. K.; Colombo, L. Electronics Based on Two-Dimensional Materials. *Nat. Nanotechnol.* **2014**, *9*, 768–779.
(4) Radisavljevic, B.; Radenovic, A.; Brivio, J.; Giacometti, V.; Kis, A. Single-Layer MoS_2 Transistors. *Nat. Nanotechnol.* **2011**, *6*, 147–150.
(5) Li, X.; Yang, L.; Si, M.; Li, S.; Huang, M.; Ye, P.; Wu, Y. Performance Potential and Limit of MoS_2 Transistors. *Adv. Mater.* **2015**, *27*, 1547–1552.
(6) Li, X.; Xiong, X.; Li, T.; Li, S.; Zhang, Z.; Wu, Y. Effect of Dielectric Interface on the Performance of MoS_2 Transistors. *ACS Appl. Mater. Interfaces* **2017**, *9*, 44602–44608.
(7) Li, X.; Li, T.; Zhang, Z.; Xiong, X.; Li, S.; Wu, Y. Tunable Low-Frequency Noise in Dual-Gate MoS_2 Transistors. *IEEE Electron Device Lett.* **2018**, *39*, 131–134.
(8) Krasnozhon, D.; Lembke, D.; Nyffeler, C.; Leblebici, Y.; Kis, A. MoS_2 Transistors Operating at Gigahertz Frequencies. *Nano Lett.* **2014**, *14*, 5905–5911.
(9) Lopez-Sanchez, O.; Lembke, D.; Kayci, M.; Radenovic, A.; Kis, A. Ultrasensitive Photodetectors Based on Monolayer MoS_2. *Nat. Nanotechnol.* **2013**, *8*, 497–501.
(10) Late, D. J.; Huang, Y.-K.; Liu, B.; Acharya, J.; Shirodkar, S. N.; Luo, J.; Yan, A.; Charles, D.; Waghmare, U. V.; Dravid, V. P.; Rao, C. N. R. Sensing Behavior of Atomically Thin-Layered MoS_2 Transistors. *ACS Nano* **2013**, *7*, 4879–4891.
(11) Bertolazzi, S.; Krasnozhon, D.; Kis, A. Nonvolatile Memory Cells Based on MoS_2/Graphene Heterostructures. *ACS Nano* **2013**, *7*, 3246–3252.
(12) Wang, H.; Yu, L.; Lee, Y.-H.; Shi, Y.; Hsu, A.; Chin, M. L.; Li, L.-J.; Dubey, M.; Kong, J.; Palacios, T. Integrated Circuits Based on Bilayer MoS_2 Transistors. *Nano Lett.* **2012**, *12*, 4674–4680.
(13) Liu, H.; Neal, A. T.; Ye, P. D. Channel Length Scaling of MoS_2 MOSFETs. *ACS Nano* **2012**, *6*, 8563–8569.
(14) Li, X.; Gao, T.; Wu, Y. Development of Two-Dimensional Materials for Electronic Applications. *Sci. China Inf. Sci.* **2016**, *59*, No. 061405.
(15) Li, L.; Yu, Y.; Ye, G. J.; Ge, Q.; Ou, X.; Wu, H.; Feng, D.; Chen, X. H.; Zhang, Y. Black Phosphorus Field-Effect Transistors. *Nat. Nanotechnol.* **2014**, *9*, 372–377.
(16) Xuefei, L.; Xiong, X.; Yanqing, W. Toward High-Performance Two-Dimensional Black Phosphorus Electronic and Optoelectronic Devices. *Chin. Phys. B* **2017**, *26*, No. 037307.
(17) Liu, H.; Neal, A. T.; Zhu, Z.; Luo, Z.; Xu, X.; Tománek, D.; Ye, P. D. Phosphorene: An Unexplored 2D Semiconductor with a High Hole Mobility. *ACS Nano* **2014**, *8*, 4033–4041.
(18) Xia, F.; Wang, H.; Jia, Y. Rediscovering Black Phosphorus as An Anisotropic Layered Material for Optoelectronics and Electronics. *Nat. Commun.* **2014**, *5*, No. 4458.
(19) Gao, T.; Li, X.; Xiong, X.; Huang, M.; Li, T.; Wu, Y. Optimized Transport Properties in Lithium Doped Black Phosphorus Transistors. *IEEE Electron Device Lett.* **2018**, *39*, 769–772.
(20) Li, X.; Du, Y.; Si, M.; Yang, L.; Li, S.; Li, T.; Xiong, X.; Ye, P.; Wu, Y. Mechanisms of Current Fluctuation in Ambipolar Black Phosphorus Field-Effect Transistors. *Nanoscale* **2016**, *8*, 3572–3578.
(21) Xiong, X.; Li, X.; Huang, M.; Li, T.; Gao, T.; Wu, Y. High Performance Black Phosphorus Electronic and Photonic Devices with HfLaO Dielectric. *IEEE Electron Device Lett.* **2018**, *39*, 127–130.
(22) Li, T.; Tian, M.; Li, S.; Huang, M.; Xiong, X.; Hu, Q.; Li, S.; Li, X.; Wu, Y. Black Phosphorus Radio Frequency Electronics at Cryogenic Temperatures. *Adv. Electron. Mater.* **2018**, *4*, No. 1800138.
(23) Wang, H.; Wang, X.; Xia, F.; Wang, L.; Jiang, H.; Xia, Q.; Chin, M. L.; Dubey, M.; Han, S.-j. Black Phosphorus Radio-Frequency Transistors. *Nano Lett.* **2014**, *14*, 6424–6429.
(24) Buscema, M.; Groenendijk, D. J.; Blanter, S. I.; Steele, G. A.; van der Zant, H. S. J.; Castellanos-Gomez, A. Fast and Broadband Photoresponse of Few-Layer Black Phosphorus Field-Effect Transistors. *Nano Lett.* **2014**, *14*, 3347–3352.
(25) Huang, M.; Wang, M.; Chen, C.; Ma, Z.; Li, X.; Han, J.; Wu, Y. Broadband Black-Phosphorus Photodetectors with High Responsivity. *Adv. Mater.* **2016**, *28*, 3481–3485.
(26) Ling, X.; Wang, H.; Huang, S.; Xia, F.; Dresselhaus, M. S. The Renaissance of Black Phosphorus. *Proc. Natl. Acad. Sci. U.S.A.* **2015**, *112*, 4523–4530.
(27) Lee, S.; Yang, F.; Suh, J.; Yang, S.; Lee, Y.; Li, G.; Sung Choe, H.; Suslu, A.; Chen, Y.; Ko, C.; Park, J.; Liu, K.; Li, J.; Hippalgaonkar, K.; Urban, J. J.; Tongay, S.; Wu, J. Anisotropic in-Plane Thermal Conductivity of Black Phosphorus Nanoribbons at Temperatures Higher than 100 K. *Nat. Commun.* **2015**, *6*, No. 8573.
(28) Luo, Z.; Maassen, J.; Deng, Y.; Du, Y.; Garrelts, R. P.; Lundstrom, M. S.; Ye, P. D.; Xu, X. Anisotropic in-Plane Thermal Conductivity Observed in Few-Layer Black Phosphorus. *Nat. Commun.* **2015**, *6*, No. 8572.
(29) Wang, X.; Jones, A. M.; Seyler, K. L.; Tran, V.; Jia, Y.; Zhao, H.; Wang, H.; Yang, L.; Xu, X.; Xia, F. Highly Anisotropic and Robust Excitons in Monolayer Black Phosphorus. *Nat. Nanotechnol.* **2015**, *10*, 517–521.
(30) Hong, T.; Chamlagain, B.; Lin, W.; Chuang, H.-J.; Pan, M.; Zhou, Z.; Xu, Y.-Q. Polarized Photocurrent Response in Black Phosphorus Field-Effect Transistors. *Nanoscale* **2014**, *6*, 8978–8983.
(31) Zhu, W.; Yogeesh, M. N.; Yang, S.; Aldave, S. H.; Kim, J.-S.; Sonde, S.; Tao, L.; Lu, N.; Akinwande, D. Flexible Black Phosphorus Ambipolar Transistors, Circuits and AM Demodulator. *Nano Lett.* **2015**, *15*, 1883–1890.
(32) Haratipour, N.; Robbins, M. C.; Koester, S. J. Black Phosphorus p-MOSFETs With 7-nm HfO_2 Gate Dielectric and Low Contact Resistance. *IEEE Electron Device Lett.* **2015**, *36*, 411–413.
(33) Joshua, O. I.; Gary, A. S.; Herre, S. J. vdZ.; Andres, C.-G. Environmental Instability of Few-Layer Black Phosphorus. *2D Mater.* **2015**, *2*, No. 011002.
(34) Yau, S.-L.; Moffat, T. P.; Bard, A. J.; Zhang, Z.; Lerner, M. M. STM of the (010) Surface of Orthorhombic Phosphorus. *Chem. Phys. Lett.* **1992**, *198*, 383–388.
(35) Zhu, X.; Zhang, T.; Jiang, D.; Duan, H.; Sun, Z.; Zhang, M.; Jin, H.; Guan, R.; Liu, Y.; Chen, M.; Ji, H.; Du, P.; Yan, W.; Wei, S.; Lu, Y.; Yang, S. Stabilizing Black Phosphorus Nanosheets via Edge-Selective Bonding of Sacrificial C60 Molecules. *Nat. Commun.* **2018**, *9*, No. 4177.
(36) Yang, B.; Wan, B.; Zhou, Q.; Wang, Y.; Hu, W.; Lv, W.; Chen, Q.; Zeng, Z.; Wen, F.; Xiang, J.; Yuan, S.; Wang, J.; Zhang, B.; Wang, W.; Zhang, J.; Xu, B.; Zhao, Z.; Tian, Y.; Liu, Z. Te-Doped Black Phosphorus Field-Effect Transistors. *Adv. Mater.* **2016**, *28*, 9408–9415.
(37) Wood, J. D.; Wells, S. A.; Jariwala, D.; Chen, K.-S.; Cho, E.; Sangwan, V. K.; Liu, X.; Lauhon, L. J.; Marks, T. J.; Hersam, M. C. Effective Passivation of Exfoliated Black Phosphorus Transistors against Ambient Degradation. *Nano Lett.* **2014**, *14*, 6964–6970.
(38) Liu, H.; Neal, A. T.; Si, M.; Du, Y.; Ye, P. D. The Effect of Dielectric Capping on Few-Layer Phosphorene Transistors: Tuning the Schottky Barrier Heights. *IEEE Electron Device Lett.* **2014**, *35*, 795–797.
(39) Deng, B.; Tran, V.; Xie, Y.; Jiang, H.; Li, C.; Guo, Q.; Wang, X.; Tian, H.; Koester, S. J.; Wang, H.; Cha, J. J.; Xia, Q.; Yang, L.; Xia, F. Efficient Electrical Control of Thin-Film Black Phosphorus Bandgap. *Nat. Commun.* **2017**, *8*, No. 14474.

DOI: 10.1021/acsami.8b16507
ACS Appl. Mater. Interfaces 2019, 11, 1587–1594

(40) Du, Y.; Liu, H.; Deng, Y.; Ye, P. D. Device Perspective for Black Phosphorus Field-Effect Transistors: Contact Resistance, Ambipolar Behavior, and Scaling. *ACS Nano* **2014**, *8*, 10035−10042.
(41) Kim, J. S.; Jeon, P. J.; Lee, J.; Choi, K.; Lee, H. S.; Cho, Y.; Lee, Y. T.; Hwang, D. K.; Im, S. Dual Gate Black Phosphorus Field Effect Transistors on Glass for NOR Logic and Organic Light Emitting Diode Switching. *Nano Lett.* **2015**, *15*, 5778−5783.
(42) Wang, L.; Li, C.; Ryan, A. J.; Armes, S. P. Synthesis and Peptide-Induced Degradation of Biocompatible Fibers Based on Highly Branched Poly(2-hydroxyethyl methacrylate). *Adv. Mater.* **2006**, *18*, 1566−1570.
(43) Johnson, R. P.; Jeong, Y.-I.; Choi, E.; Chung, C.-W.; Kang, D. H.; Oh, S.-O.; Suh, H.; Kim, I. Biocompatible Poly(2-hydroxyethyl methacrylate)-b-poly(L-histidine) Hybrid Materials for pH-Sensitive Intracellular Anticancer Drug Delivery. *Adv. Funct. Mater.* **2012**, *22*, 1058−1068.
(44) Braunecker, W. A.; Matyjaszewski, K. Controlled/Living Radical Polymerization: Features, Developments, and Perspectives. *Prog. Polym. Sci.* **2007**, *32*, 93−146.
(45) Li, L.; Engel, M.; Farmer, D. B.; Han, S.-j.; Wong, H. S. P. High-Performance p-Type Black Phosphorus Transistor with Scandium Contact. *ACS Nano* **2016**, *10*, 4672−4677.
(46) Li, X.; Grassi, R.; Li, S.; Li, T.; Xiong, X.; Low, T.; Wu, Y. Anomalous Temperature Dependence in Metal-Black Phosphorus Contact. *Nano Lett.* **2018**, *18*, 26−31.
(47) Doganov, R. A.; O'Farrell, E. C. T.; Koenig, S. P.; Yeo, Y.; Ziletti, A.; Carvalho, A.; Campbell, D. K.; Coker, D. F.; Watanabe, K.; Taniguchi, T.; Neto, A. H. C.; Özyilmaz, B. Transport Properties of Pristine Few-Layer Black Phosphorus by van der Waals Passivation in an Inert Atmosphere. *Nat. Commun.* **2015**, *6*, No. 6647.
(48) Kim, J.-S.; Liu, Y.; Zhu, W.; Kim, S.; Wu, D.; Tao, L.; Dodabalapur, A.; Lai, K.; Akinwande, D. Toward Air-Stable Multilayer Phosphorene Thin-Films and Transistors. *Sci. Rep.* **2015**, *5*, No. 8989.
(49) Ahmed, F.; Kim, Y. D.; Choi, M. S.; Liu, X.; Qu, D.; Yang, Z.; Hu, J.; Herman, I. P.; Hone, J.; Yoo, W. J. High Electric Field Carrier Transport and Power Dissipation in Multilayer Black Phosphorus Field Effect Transistor with Dielectric Engineering. *Adv. Funct. Mater.* **2016**, *27*, No. 1604025.
(50) Li, T.; Zhang, Z.; Li, X.; Huang, M.; Li, S.; Li, S.; Wu, Y. High Field Transport of High Performance Black Phosphorus Transistors. *Appl. Phys. Lett.* **2017**, *110*, No. 163507.
(51) Si, M.; Yang, L.; Du, Y.; Ye, P. D. In *Black Phosphorus Field-Effect Transistor with Record Drain Current Exceeding 1 A/mm*, 2017 75th Annual Device Research Conference (DRC), June 25−28, 2017; pp 1−2.
(52) Kang, J.; Jariwala, D.; Ryder, C. R.; Wells, S. A.; Choi, Y.; Hwang, E.; Cho, J. H.; Marks, T. J.; Hersam, M. C. Probing Out-of-Plane Charge Transport in Black Phosphorus with Graphene-Contacted Vertical Field-Effect Transistors. *Nano Lett.* **2016**, *16*, 2580−2585.

DOI: 10.1021/acsami.8b16507
ACS Appl. Mater. Interfaces 2019, 11, 1587−1594